

■ Scientific Justification (required for all)

Does the dust around a changing nucleus keep up with it? An active galactic nucleus (AGN) can undergo large luminosity changes within years [1,2], sometimes accompanied by a transformation of its continuum and broad-line spectrum [3–5]. Its surrounding dust records these changes through delayed heating and through destruction or replenishment of the emitting material. Individual nuclei show that dust boundaries evolve over years [18,19]; the next step is to establish how the response depends on the direction and history of nuclear activity. Changing-look and strongly variable AGN followed through rises, fades and recoveries provide this experiment.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$: 12 fading and 12 rising sources**. Every target has ZTF/WISE light curves, SPHEREx spectra and multi-epoch optical spectroscopy, including 2026 Palomar/NGPS spectra. Together, their recorded histories provide ensemble coverage of different stages of nuclear activity and dust response. MIRI supplies the essential warm continuum, both silicate features and spatial constraints on host emission, connecting each history to its warm-dust spectrum. The programme addresses two linked questions:

1. How strongly does the warm spectrum remember the nuclear history? Delayed heating predicts a warm excess after a decline and a deficit after a rise [6,7]. Twelve rising and twelve fading selection histories, each tied to its own 2010 WISE measurement, test the amplitude and persistence of this spectral memory.

2. Does the dust boundary move, or only its temperature? Delayed heating of unchanged dust is the null model. The silicate features and the full 5–28 μm continuum test whether the spectra require changes in the dust distribution beyond this response [6,7,18,19].

From infrared echoes to dust evolution. WISE observations have established infrared echoes of changing-look transitions [8–11], supporting intrinsic changes in nuclear power in those sources. Broadband colours, however, combine temperature, emitting area, extinction and host light [12–14]. The proposed spectra test the warm spectral signature of those histories and constrain which properties of the emitting material produce it.

The equilibrium sublimation radius scales approximately as $L^{1/2}$ [15–17], but the physical dust boundary can adjust over years [18]. Individual nuclei already suggest different responses: the smaller dust radius after Mrk 590 faded supports replenishment [19], whereas the temperature variations in Mrk 817 favour heating and cooling at a nearly fixed boundary [20]. These results motivate a comparison across both directions of change. Recovering sources further test whether the spectrum depends on the path to its present state.

MIRI’s diagnostic power. SPHEREx covers 0.75–5 μm [21], constraining hot dust close to the nucleus [20,22]. Longer delays and weaker variability at longer wavelengths [6,23,24] make the warm continuum sensitive to earlier illumination. MRS supplies the essential 4.9–27.9 μm spectrum [25]: the continuum measures the imprint, the paired silicate features constrain geometry and optical depth [26,27], and spatial profiles and aromatic bands constrain host emission [14,28,29]. MIRI extends the NEOWISE/SPHEREx record into the warm-dust regime and connects it to the 2010 W3/W4 baseline. This combination tests how dust at different temperatures responds to the same recorded activity.

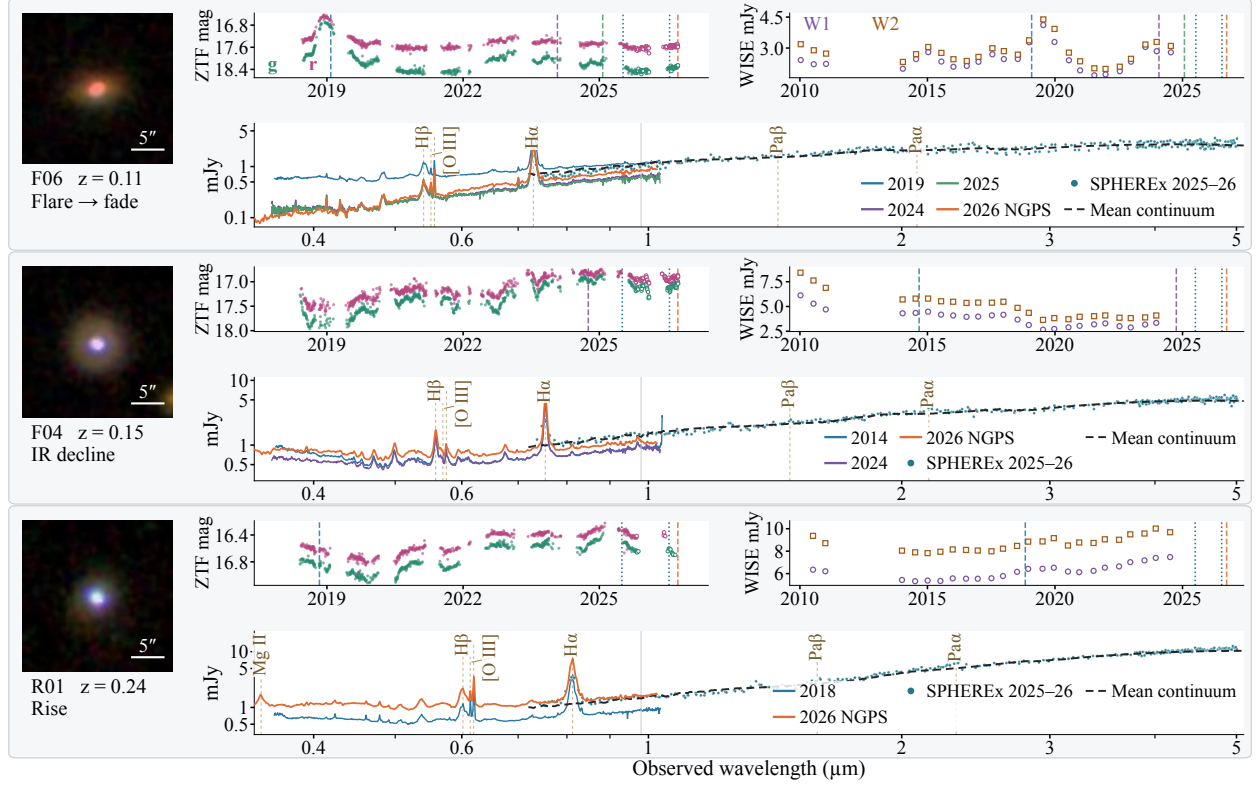


Figure 1: **Three recorded histories connect nuclear activity to hot dust.** F06: a 2018–2019 flare followed by fading; F04: a long infrared decline with modest optical recovery; R01: rising light curves and optical spectra. Each block combines an SDSS image (5'' bar), ZTF g/r and WISE W1/W2 histories, and archival/2026 NGPS spectra with SPHEREx. Open ZTF points extend the record with alert photometry to 2026 September; coloured ticks mark spectral epochs. Dashed curves show the mean host+disc+blackbody continuum fitted to SPHEREx ($T_{\text{col}} \approx 1100\text{--}1200$ K, model-dependent). Gold markers identify selected expected lines. Wavelengths are observed-frame; spectral fluxes are in mJy, without inter-epoch rescaling.

Tracking changes in AGN activity. We combine catalogued changing-look AGN [32–35] with large-amplitude ZTF/WISE variables [5,36], emphasizing variability direction, amplitude and timing across spectral classifications [33,37]. A variability manifold groups similar light-curve shapes to identify candidates [38]; multi-epoch spectra establish continuum and broad-line changes. The 12 rising and 12 fading selection histories span sustained changes, plateaus and recoveries. Each target has approximately a decade of W1/W2 monitoring; W1 95th-to-5th-percentile flux ratios span 1.3–2.3 before host subtraction. **This diversity tests infrared memory across different stages of change.** Recovering sources test whether nuclei at comparable current power retain different warm spectra because they followed different paths. SPHEREx and the 2026 NGPS spectra connect these histories to hot-dust emission and optical spectral state (Figure 1).

Measuring the spectral memory of accretion. Each target’s 2010 W3/W4 photometry provides a historical reference for its MIRI spectrum. We will fit both epochs through their instrument bandpasses, with a shared host component, to measure the change in nuclear warm emission. This comparison anchors the model normalization to the same object. The MRS rest-frame 8–13 μm continuum and silicate profiles then test its spectral response to the full recorded history (Figure 2). Sustained rises and declines test the expected deficit and excess relative to instantaneous response; recoveries test contributions from successive activity phases. Optical decomposition, WISE and SPHEREx constrain the nuclear illumination and hot-dust response [4,20,22]. The warm continuum and silicates constrain temperature and geometry, which set the model’s radii and delays [26,28]; spatial profiles and PAH bands constrain host emission [14,29]. Optical monitoring near JWST anchors the current nuclear state. Event dates, gaps and the optical-to-heating conversion enter the joint fit.

Forecast. Optically thin shells [45] driven by a host-subtracted W1 heating proxy provide a quantitative planning case for all 24 recorded histories, using monochromatic observed-frame 12 μm flux as an approximation to W3. The sign-weighted mean separation is 0.062 dex for delayed versus instantaneous heating and 0.051 dex for evolving versus fixed dust. Under the adopted error model, the mean error is 0.014 dex (4.5 and 3.7σ). The 0.03–0.05 dex photometric/decomposition term grows to 0.03–0.19 dex including the current W1 host-fraction term. Twelve targets exceed 3σ separation for at least one model pair with measurement errors alone; five do so including that host term. These forecasts assume constant illumination outside the recorded history and a fixed shell geometry. Opposite response signs suppress common calibration shifts. Recovery simulations will extend this calculation to bandpass-integrated spectra, allowed histories and geometries, and correlated uncertainties.

Testing for dust evolution. We will use clumpy and disc-plus-wind geometries [40–42] to construct responses to the reconstructed histories [6,7], allowing for composition and grain-size uncertainty [43]. Fixed distributions change temperature; evolving distributions also change the amount or location of dust [18,19,22]. Both include light-travel delays. The diagnostic $\delta_{\text{warm}} = \log_{10}[L_{8-13\mu\text{m}}^{\text{MRS}}/L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})]$ measures the warm-continuum residual relative to a *delayed fixed-distribution* prediction. Coherent continuum and silicate residuals across the allowed fixed models test the need for structural evolution [28], with host and calibration uncertainties carried through the fit [44]. The forecast uses the limiting case of immediate boundary adjustment after illumination arrives; the science analysis will test finite adjustment times across the ensemble’s event ages.

The historical comparison measures the warm response even where both dust models fit, and quantifies its impact on infrared estimates of nuclear power and covering factor [30,31].

Complementary tests of obscuration and excitation. The paired silicate profiles constrain optical depth and geometry [26,27]. Combined with the optical history and high-ionization lines, they test intrinsic dimming against obscuration. [Ne VI] 7.7 and [Ne V] 14.3 μm provide excitation diagnostics throughout the sample [50]; the nominal MRS range also includes [Ne V] 24.3 μm for 13 targets and [O IV] 25.9 μm for nine. Spatially resolved PAH and H_2 emission characterizes the hosts of both cohorts [29,49].

Recorded histories → MIRI spectra → dust physics

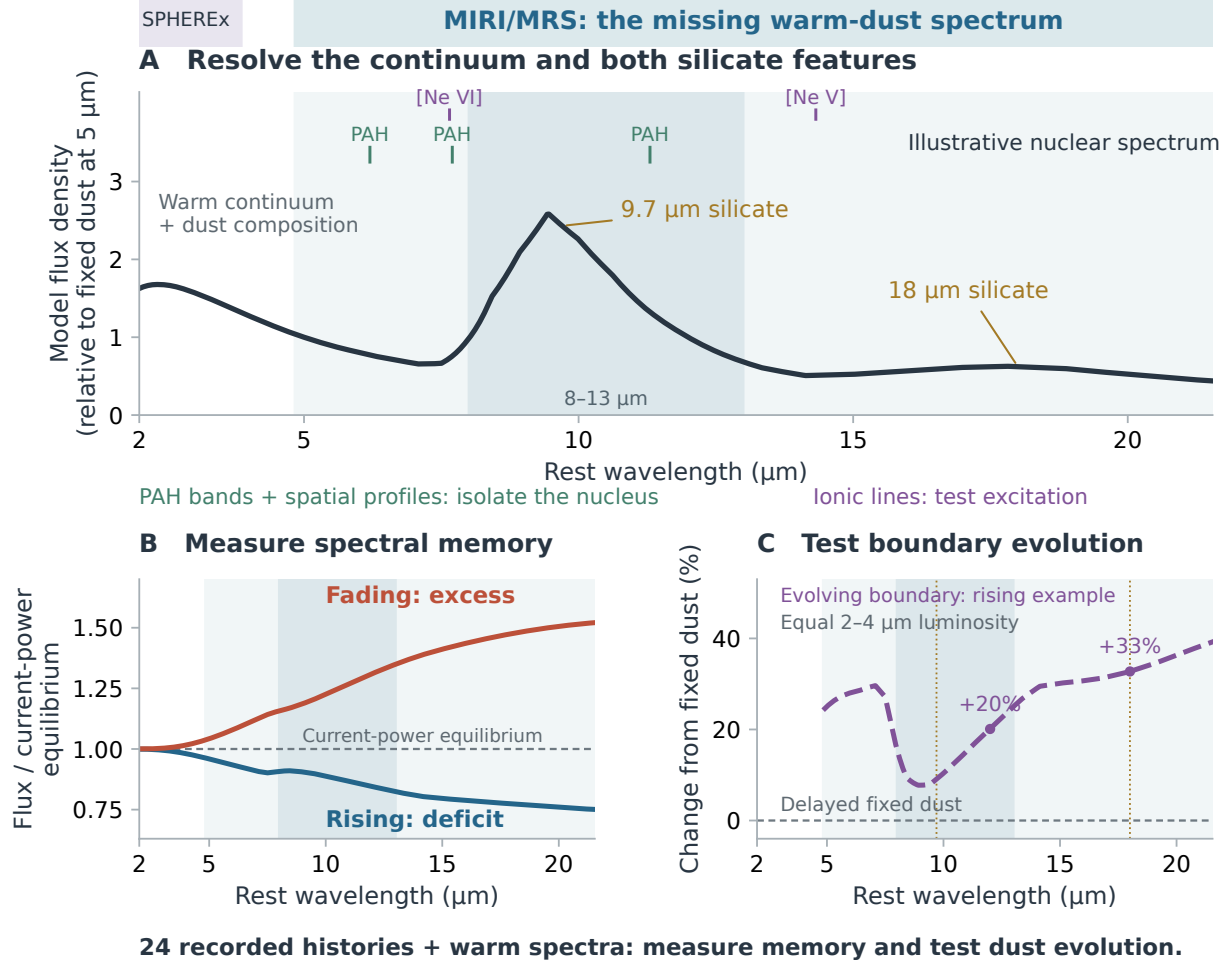


Figure 2: **The histories are in hand; MIRI supplies the decisive warm spectrum.** A: an illustrative nuclear spectrum locates the warm continuum and paired silicate features that MIRI adds to the hot-dust record. PAH and ionic ticks mark diagnostic locations, not simulated line emission. B: delayed heating produces a warm excess after fading and a deficit after rising, relative to current-power equilibrium. C: the rising example isolates the additional boundary-evolution signature after matching the integrated 2–4 μm luminosity, $100(F_V^{\text{evolving}}/F_V^{\text{fixed}} - 1)$; zero denotes delayed fixed dust. These illustrative shell calculations use fourfold luminosity changes at ten initial inner-radius light-crossing times and silicate/graphite opacities [6,7,45]. The evolving boundary adjusts as changed illumination arrives. Shading marks MRS coverage common to all 24 targets and the primary 8–13 μm band; the SPHEREx strip shows its common coverage within the plotted range. Spatial profiles and PAH bands constrain the host; ionic lines test excitation [25,28,29,50].

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. We request one MIRI/MRS science visit and a matching sky visit per target. All three grating settings, nuclear target acquisition and a four-point dither provide approximately 4.9–27.9 μm coverage [25,46]. Both silicate peaks fall within this range for all 24 targets. The primary rest-frame 8–13 μm continuum diagnostic is accessible throughout the sample without channel 4C; the full spectrum constrains the longer-wavelength continuum and 18 μm profile.

Archival total-source W3 and W4 fluxes span 1.9–48.4 and 6.4–82.6 mJy. We target continuum S/N of 50 per $R = 100$ bin near rest 5 and 12 μm and 30 around the 18 μm feature, corresponding to 2% and 3.3% statistical errors per bin. We use FULL/FASTR1, one integration per dither, and 27/60/27 groups in settings A/B/C for 23 targets. Four dithers give 299.7/666.0/299.7 s on source, with all four channels observed simultaneously. The W3-faint J083826.50+371906.7 uses 80 groups in B (888.0 s). Equal-depth sky exposures preserve extended host emission for decomposition [46]. Science-plus-sky integration totals 17.00 h. **We request 67.0 h**, including acquisition, visit operations, slews and observatory overheads calculated with APT 2026.5.1: a Medium programme [47].

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandeia 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_{\nu} \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures and extraction radii of 0.3''/0.6''/0.9''. They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts giving S/N above 50 without saturation over the adopted flux ranges [48].

Connecting the observing epochs. The historical baseline gives each MIRI spectrum its temporal context, with 2010 AllWISE W3/W4 photometry providing the warm-emission reference. WISE and SPHEREx constrain the dust response at their observed epochs; optical spectroscopy tracks the nuclear spectrum. Optical monitoring near JWST anchors the nuclear continuum at the MIRI epoch. One MIRI visit per target connects recorded activity to the warm-dust response across 24 histories spanning different directions, amplitudes and event ages.

Separating nuclear dust from the host. Each MRS cube measures the nucleus and surrounding host together. Wavelength-dependent point-spread-function fitting [51] separates unresolved and extended emission [28]; surrounding spaxels constrain the host spectrum directly. Joint modelling distinguishes continuum, silicate, PAH and H_2 contributions [13,49]. The 6.2, 7.7 and 11.3 μm PAH bands constrain star formation with allowance for AGN processing [29]; [Ne v] and [Ne vi] trace highly ionised gas [50]. Alternative extractions and host templates [14] define conservative nuclear-flux bounds.

We will compare the 2010 W3/W4 photometry with redshifted nuclear-plus-host models integrated through the WISE bandpasses, including colour corrections and emission outside MRS coverage. Archival imaging constrains host light beyond the MRS field and within the SPHEREx aperture. Both science tests propagate nuclear-flux bounds and wavelength-dependent calibration covariance [44].

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [46]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

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